

From Laplace to Fourier



Portrait of Pierre-Simon Marquis de Laplace, 1749–1827



Portrait of Joe Fourier, 1768–1830

The Laplace transform can be used to simplify solving some differential equations. The transform essentially turns differential equations into algebraic equations. It's well-covered in our textbook here: https://www.jirka.org/diffyqs/html/laplace_section.html. Our textbook also covers the Fourier transform in the context of partial differential equations in this chapter: https://www.jirka.org/diffyqs/html/FS_chapter.html. You might not realize it from reading our book, but the Laplace and Fourier transforms are closely related—in fact, the Fourier transform *is* a Laplace transform with some restrictions applied.

These notes briefly:

1. Introduce the Laplace transform and show a few ODE examples.
2. Illustrate a few important properties of the Laplace transform.
3. Introduce the Fourier transform as a special case of the Laplace transform.

The Laplace Transform

Think of the Laplace transform $\mathcal{L}$ as a kind of function that takes one function f as input and returns another function F as output. It's defined as:

$$\mathcal{L}(f(t)) = F(s) = \int_0^{\infty} e^{-st} f(t) dt. \quad (1)$$

The transform multiplies the input function f by e^{-st} and integrates from zero to infinity. And, the parameter s of the new function is complex-valued (real and imaginary components). This seems like it makes things more complicated!

Although it seems complicated, in practice the Laplace transform can dramatically simplify solving many differential equations—in **particular ones that are not homogeneous**. Transforms are rarely integrated by hand. Usually we simply look up the transforms of functions from a table or using software. Table 6.1 of our book lists some common transforms. The transforms in the table are all invertible.

Example using the Laplace transform to solve an ODE

The ode in this example is related to problem 1.4.9 in our book—water flowing through two reservoirs. In particular, our ODE shows flow in the lower reservoir under some simplifying assumptions. It's a linear first-order, constant coefficient, non-homogeneous, ordinary differential equation:

$$f'(t) + rf(t) = rqe^{-rt}, \quad f(0) = 0,$$

for constants r, q and time t . There are many possible ways to solve for $f(t)$, including using an integrating factor or the Laplace transform. Let's use the Laplace transform! We'll use capital $F(s)$ to be the transform of $f(t)$ to get:

$$sF(s) - f(0) + rF(s) = \frac{rq}{s+r},$$

and since $f(0) = 0$, then:

$$sF(s) + rF(s) = \frac{rq}{s+r}.$$

Now we just solve for F and then apply the inverse Laplace transform to get f !

$$F(s) = \frac{rq}{(s+r)^2}.$$

Applying the inverse Laplace transform yields:

$$f(t) = rqte^{-rt}.$$

Properties of the Laplace transform

Linearity

The Laplace transform is just an integral and inherits linearity from that. In particular,

$$\mathcal{L}(f+g) = \mathcal{L}(f) + \mathcal{L}(g) = F(s) + G(s).$$

Time shifting

$$\mathcal{L}(f(t-T)) = e^{-sT}F(s).$$

Convolution

A *convolution* of two functions f and g is defined to be:

$$(f * g)(t) = \int_0^t f(u)g(t-u)du.$$

This may seem like an odd operation but occurs naturally in many settings. In many ways, convolution has product-like properties:

$$f * g = g * f$$

$$(f + g) * h = f * h + g * h.$$

$$(f * g) * h = f * (g * h).$$

Importantly, the Laplace transform of a convolution is the product of the Laplace transforms:

$$\mathcal{L}(f * g) = F(s)G(s).$$

The upshot is that if you ever have a product of transformed functions (like $F(s)G(s)$ above), then the solution will be the *convolution* of their inverse transforms.

Here is how that works. Say you wish to find the inverse Laplace transform of $\frac{1}{s^2} \frac{1}{s+1}$:

1. Look up the inverse transform of $\frac{1}{s^2} = t$.
2. Look up the inverse transform of $\frac{1}{s+1} = e^{-t}$.
3. Now you know that the solution is the convolution of t with e^{-t} , or $\int_0^t u e^{-(t-u)} du$.
4. Finish by evaluating $\int_0^t u e^{-(t-u)} du = e^{-t} + t - 1$ (for instance with integration by parts!).

Dirac's delta function

Named for Paul Dirac, the delta function, or δ is not exactly a normal function. Imagine a box with area one centered at 0 on the x -axis. Such a box between, say, $-\frac{1}{2} \leq x \leq \frac{1}{2}$ would have height 1. But between $-\frac{1}{4} \leq x \leq \frac{1}{4}$ the height of the box would have to be 2 to still have area equal to 1. Think of the delta function as the limit of those boxes as the x -values approach zero. The function is essentially zero everywhere and infinite when $x = 0$.

The Laplace transform of the delta function is 1, $\mathcal{L}(\delta(t)) = 1$. Solving a differential equation for *impulse response* means setting the ODE equal to $\delta(t)$. See Example 6.4.2 for a good example.

From Laplace to Fourier...

You may have noticed that integration in the Laplace transform is usually defined from 0 to ∞ . It's also possible to define a bi-infinite (also called bi-lateral) transform:

$$\mathcal{L}(f(t)) = F(s) = \int_{-\infty}^{\infty} e^{-st} f(t) dt. \quad (2)$$

This version of the transform shares many of the same properties with Equation(1). It accounts for the entire history of a signal, not just from time greater than zero.

Earlier we mentioned that the parameter s of the transformed function is complex-valued—so it looks like $\alpha + \beta i$ where $i = \sqrt{-1}$ is the imaginary unit. The real part α acts as a “damping” or “growth” factor. The imaginary part β deals with oscillation. If we restrict that parameter so the real part $\alpha = 0$, then $s = \beta i$ is a purely imaginary number and the transform is focused on steady-state oscillation. That's the Fourier transform! Thus, the Fourier transform is a special case of the Laplace transform.

Finally, assume that the function f is periodic with period T —that is the function repeats itself over interval length T . For the transform to be consistent with this periodicity, the kernel e^{-st} must satisfy the same boundary conditions

as the function itself. Since $f(t) = f(t + T)$, that means that $e^{-st} = e^{-s(t+T)} = e^{-st}e^{-sT}$. That can only be true if $e^{-sT} = 1$, which means that $-sT$ is a multiple of 2π because of Euler's formula: $e^{\beta ti} = \cos(\beta t) + i \sin(\beta t)$. Therefore,

$$-sT = -i(2\pi n) \quad \text{for } n = 0, \pm 1, \pm 2, \dots$$

Solving for s shows that rather than a continuous line along the imaginary axis, s takes on discrete points:

$$s_n = i \left(\frac{2\pi n}{T} \right).$$

That means that the corresponding Fourier transform is

$$F(s_n) = \int_{-\infty}^{\infty} e^{-it \frac{2\pi n}{T}} f(t) dt. \quad (3)$$

The Fourier Series

Notice that Equation(3) is defined on discrete points along the imaginary (i) axis. With some effort we will show in future notes that the inverse transform associated with Equation(3) reduces to a series:

$$f(t) = \sum_{-\infty}^{\infty} \frac{1}{T} F_T \left(i \frac{2\pi n}{T} \right) e^{i \frac{2\pi n}{T} t}, \quad (4)$$

where

$$F_T(s_n) = \int_0^T f(t) e^{-s_n t} dt$$

is the transform of $f(t)$ over one period. This is the usual Fourier series definition of the function f . The values

$$c_n = \frac{1}{T} F_T \left(i \frac{2\pi n}{T} \right)$$

are called the Fourier coefficients of f .

The Discrete and Fast Fourier Transforms

Functions (signals) are often measured over discrete points in time to produce a finite sequence of values. The *discrete fourier transform* (DFT) converts that finite sequence into another finite sequence of the same length that represent the Fourier coefficients (frequency components) of the signal. When sampled at equally-spaced points, the DFT represents a trig polynomial that interpolates the signal at those points.

The DFT is typically computed using the *fast fourier transform* (FFT), a remarkably-efficient algorithm developed by Cooley and Tukey in 1965—but later discovered to be originally invented by Gauss in 1805. The FFT and its variations is an enabling technology for much of modern signal processing. Its importance is hard to over-state.